

DPP-01

CS & IT

Algorithm

Divide and Conquer

- Q1** Consider an array containing the following elements in unsorted order (placed randomly) but 120 as first elements 120 160 30 190 14 24 70 180 110

Quick sort partitioning algorithm is applied by choosing first elements as pivot element. Then what is the total number of arrangements of array integers are possible preserving the effect of first pass of partitioning algorithm.

- (A) 680 (B) 700
(C) 720 (D) 740

- Q2** Let $T(n) = [n(\log(n^3) - \log n) + \log n]n + \log n$. complexity of $T(n)$ is

- (A) $O(n^2)$ (B) $O(n^3)$
(C) $O(n \log n)$ (D) $O(n^2 \log n)$

- Q3** Assume that there are 4 sorted lists of $\frac{n}{4}$ elements each, if these lists are merged into a single sorted list of 'n' elements then how many key comparisons are required in the worst case using an efficient algorithm?

- (A) $2n - 3$ (B) $\frac{7}{4}n - 3$
(C) $\frac{9}{4}n - 3$ (D) $\frac{6}{4}n - 3$

- Q4** Consider the number in the sequence 2 5 11 17 19 21 26 33 39 40 51 65 79 88 99 Using binary search, the number of comparisons required to search elements '2' is ____

- Q5** Merging 4 sorted files having 400, 100, 250, 50 records will take how many comparisons to be merged into a single sorted file, if 2 files are merged at a time?

- (A) 800 (B) 799
(C) 1597 (D) 1347

- Q6** Consider a machine which needs a minimum of 50 seconds to sort 500 names by quick sort, then what is the minimum time required to sort 50 names (approximately) is ____ (round off to 2 decimal & Consider log with base 2)

- Q7** What is the total number of comparisons that will be required in worst case to merge the following sorted files into a single sorted file by merging together two files at a time ____.

Files	F ₁	F ₂	F ₃	F ₄
Number of records	40	42	44	46



Answer Key

Q1 C
Q2 D
Q3 A
Q4 4

Q5 D
Q6 3.14
Q7 341

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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

We have to choose the 1st elements as pivot.

Here 120 is the 1st element. After the 1st pass 1st elements goes to its exact location.

All the elements greater than 120 goes to right of 120 and lesser elements goes to left side of 120 after 1st pass.

110 30 14 24 70 120 160 180 190

5!

3!

$5! \times 3!$

720 possible arrangement

C is correct option.

Q2 Text Solution:

Given

$$T(n) = [n(\log(n^3) - \log n) + \log n] n + \log n$$

$$= \left[n \left(\log \frac{n^3}{n} \right) + \log n \right] n + \log n$$

$$= [n \log n^2 + \log n] n + \log n$$

$$= n^2 \cdot 2 \log n + n \log n + \log n$$

$$= 2n^2 \log n + n \log n + \log n, \text{ here}$$

$n^2 \log n$ is dominating term

$$= O(n^2 \log n)$$

option (d) is correct.

Q3 Text Solution:

$$\frac{n}{4} \quad \frac{n}{4}$$

$$\frac{n}{4} \quad \frac{n}{4}$$

$$\left\{ \frac{n}{2} - 1 \right\}$$

$$\left\{ \frac{n}{2} - 1 \right\}$$

Comparisons

Comparisons

$n - 1$ Comparisons

Total number of comparisons =

$$\left(\frac{n}{2} - 1 \right) + \left(\frac{n}{2} - 1 \right) + (n - 1)$$

$$= 2n - 3$$

option (a) is correct.

Q4 Text Solution:

Low = 0, high = 14

$$\text{Mid} = \frac{0+14}{2} = 7$$

$$A[7] = 33$$

$$33 > 2$$

Low = 0, high = 6;

$$\text{Mid} = \frac{0+6}{2} = 3$$

$$A[3] = 17$$

$$17 > 2$$

$$\text{Low} = 0, \text{high} = 2, \text{mid} = \frac{0+2}{2} = 1$$

$$A[1] = 5$$

$$5 > 2$$

$$\text{Low} = 0, \text{high} = 0, \text{mid} = \frac{0+0}{2} = 0$$

$$A[0] = 2$$

$$2 = 2$$

Q5 Text Solution:

Two sorted file of size m and n takes $O(m + n)$ time for merging, and $m + n - 1$ comparisons. Apply optimal merge pattern of records or files.

number of comparisons if we use two way merge pattern = 1347

Q6 Text Solution:

In best case quick sort algorithm takes $n \log n$ comparisons,

$$\text{So, } 500 \times \log_2 500 @ 4482.89$$

Which takes 50 seconds.

To sort 50 names a minimum of $50(\log_2 50) =$

282.19 comparisons

282.19 comparisons are needed.

$$\text{This takes} = 50 \times \frac{282.19}{4482.89} = 3.14 \text{ second}$$

Q7 Text Solution:

Given files

{(40, 42), 44, 46} $82 - 1 = 81$ comparisons

{82, (44, 46)} $90 - 1 = 89$ comparisons



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{82, 90} $172 - 1 = 171$ comparisons

So Total = $81 + 89 + 171 = 341$ comparisons



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